

**SIMATS ENGINEERING
ASSESSMENT TOOL 3**

COURSE NAME: Fundamentals of Data Science

COURSE CODE: DSA04

COURSE OUTCOME COVERED

CO2: Apply data pre-processing and visualization techniques using Python libraries for effective data handling. (BL4)

QUESTIONS: 05

Total Marks:100

Annexure A

ANALYTICAL PROBLEM SOLVING

Code of Conduct:

*I _____ (Name / Reg No) certify that this submission is my original work
and that I have adhered to the guidelines ~~specified for this assessment.~~
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Annexure A

Analytical Problem Solving

1. Online Food Delivery Data Analysis

An online food delivery company wants to analyze order data to improve delivery performance and customer satisfaction. Explain how:

- A. **Pandas** is used to read, clean, and filter order data
- B. **Data preprocessing** is applied to handle missing delivery time, duplicate orders, and incorrect prices
- C. **Matplotlib or Seaborn** is used to visualize delivery delays, top restaurants, and customer ratings

2. Bank Loan Approval Analysis

A bank wants to study customer loan applications and identify patterns related to loan approval and rejection. Explain how:

- A. **Pandas DataFrame operations** are used to select income, loan amount, credit score, and approval status
- B. **Preprocessing techniques** are used to handle missing income values, encode gender/employment type, and normalize loan amount
- C. **Visualization** is used to compare approved and rejected applications based on income, credit score, and loan amount

3. Smart Energy Consumption Monitoring

A smart home system collects daily electricity consumption data from different appliances such as AC, fan, light, and refrigerator. Explain how:

- A. **Pandas** is used to group electricity usage by appliance and date
- B. **Data preprocessing** is used to remove abnormal readings, fill missing values, and calculate total power consumption
- C. **Line charts and bar charts** are used to identify high-energy appliances and daily usage trends

4. Online Learning Platform Performance Analysis

An online learning platform wants to analyze student activity such as login time, video completion, quiz score, and assignment submission. Explain how:

- A. **Pandas** is used to sort, filter, and rank students based on quiz scores and completion percentage
- B. **Preprocessing** is applied to handle missing quiz scores, convert date formats, and create new columns such as performance category
- C. **Seaborn and Matplotlib** are used to visualize student engagement, score distribution, and course completion status

5. Public Transport Passenger Data Analysis

A city transport department wants to analyze bus passenger data to improve route planning and reduce overcrowding. Explain how:

- A. **Pandas** is used to analyze passenger count by route, stop, and time slot
- B. **Data preprocessing** is used to clean invalid route numbers, handle missing passenger counts, and remove duplicate records
- C. **Visualization techniques** are used to show peak travel hours, busiest routes, and passenger flow patterns

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand